
Unified Implicit Differentiation for Structured Deep Learning Layers: A Review and Extension

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Abstract

Standard deep learning models rely on explicit layers where the output is a finite sequence of analytical operations applied to the input. Recent advances have introduced *implicit layers*, where the output is defined as the solution to a mathematical problem, such as a fixed-point equation, an ordinary differential equation (ODE), or a constrained optimization problem. While these approaches appear distinct, they share a common mathematical foundation: the reliance on implicit differentiation for efficient backpropagation. In this project, we provide a unified mathematical formulation that encompasses Deep Equilibrium Models (DEQ), Neural ODEs, and Differentiable Optimization layers (OptNet). We critically review literature in this domain and propose [Your Proposed Method Name], a novel approach that improves upon existing implicit solvers by [Briefly mention your improvement, e.g., incorporating low-rank approximations/randomized numerical linear algebra]. Preliminary experiments demonstrate [Expected Results].

1 Introduction

Modern deep neural networks are predominantly built upon explicit transformations (e.g., Conv2D, BatchNorm, ReLU), where the hidden state z_{l+1} is explicitly computed from z_l . However, a growing body of research suggests that defining layers *implicitly*—via the solution to an underlying equilibrium or optimization condition—offers significant advantages in terms of memory efficiency, parameter sharing, and inductive bias.

This project focuses on the unification of these implicit methods. Specifically, we investigate the connection between Deep Equilibrium Models [4], Neural ODEs [6], and Differentiable Optimization Layers [3, 1]. By framing these diverse architectures under a single root-finding problem $g(z, x, \theta) = 0$, we leverage the Implicit Function Theorem (IFT) to derive a generalized backpropagation algorithm.

The contributions of this report are threefold:

1. We present a unified mathematical formulation for implicit layers, showing that DEQs, Neural ODEs, and OptNet are special cases of a generic root-finding problem.
2. We provide a critical review of key literature, highlighting the computational bottleneck of the Jacobian inverse computation.
3. We propose [Your Method], which [brief description of your improvement], and validate it on [Dataset/Task].

2 Unified Problem Formulation

Unlike explicit layers where $z = f(x, \theta)$, an implicit layer defines the output $z^* \in \mathbb{R}^d$ as the solution to a constraint equation. We formalize this unification below.

2.1 The Foundation of Mathematics-The Matrix and the Structured Gradient

2.1.1 Matrix Backpropagation for Deep Networks with Structured Layers

The seminal paper for matrix backpropagation [8] introduced a formal framework for training deep networks with global structured layers (e.g., SVD, eigenvalue decomposition) end-to-end.

This work established Matrix Backpropagation (MBP): A methodology to compute gradients for matrix functions using the invariant property of the Frobenius inner product (trace), generalizing the chain rule to matrix variables.

$$\begin{aligned} \frac{\partial L^{(l+1)}}{\partial Y} : dY &= \frac{\partial L^{(l+1)}}{\partial Y} : \mathcal{L}(dX) \triangleq \mathcal{L}^* \left(\frac{\partial L^{(l+1)}}{\partial Y} \right) : dX \\ &\Rightarrow \mathcal{L}^* \left(\frac{\partial L^{(l+1)}}{\partial Y} \right) = \frac{\partial L^{(l)}}{\partial X} \end{aligned} \quad (1)$$

They also applied this framework to spectral layers, and derived the specific gradients for layers involving SVD ($X = U\Sigma V^T$) and symmetric EIG ($X = U\Sigma U^T$).

2.1.2 A Unified Framework for Matrix Backpropagation

This paper [7] revisits the Ionescu approach, corrects it, and proves its theoretical equivalence to the Daleckii-Krein/Bhatia (DK/B) formula, arguing that DK/B is superior for implementation.

This work generalized DK/B Formula: For a diagonalizable matrix $X = U\Lambda U^{-1}$ and $Y = f(X)$:

$$\frac{\partial \phi}{\partial X} = U^{-T} [f^{[1]}(X) \odot (U^T \frac{\partial \phi}{\partial Y} U^{-T})] U^T$$

where

$$[f^{[1]}(X)]_{ij} = \begin{cases} \frac{f(\lambda_i) - f(\lambda_j)}{\lambda_i - \lambda_j}, & i \neq j \\ f'(\lambda_i), & \text{otherwise.} \end{cases}$$

And for the SPD Case, since $U^{-1} = U^T$, this simplifies to:

$$\frac{\partial \phi}{\partial X} = U [f^{[1]}(X) \odot (U^T \frac{\partial \phi}{\partial Y} U)] U^T$$

Darley and Bonnet (2025) provide this unified framework that corrects theoretical inaccuracies and proves the equivalence of the original method to the Daleckii-Krein/Bhatia (DK/B) formula. The Unified approach addresses the critical issue of numerical instability in Ionescu's formulation where close eigenvalues lead to gradient explosion by utilizing the DK/B formula with Loewner matrices to robustly handle degenerate cases, while simultaneously extending the theory to general diagonalizable matrices and achieving an 8-37% improvement in computational speed.

The experiments can be seen in Appendix A.

2.2 Optimization as Model: From Constraint Networks to Embedded Solvers

2.2.1 OptNet: Differentiable Quadratic Programs as Layers

Amos and Kolter [3] introduced the OptNet architecture, which integrates optimization problems specifically Quadratic Programs (QPs) as individual layers within deep networks. Unlike standard layers with explicit analytical forms, the output of an OptNet layer is the solution to a constrained optimization problem.

Formally, the layer output z_{i+1} is defined as the minimizer of a QP:

$$\begin{aligned} z_{i+1} &= \underset{z}{\operatorname{argmin}} \quad \frac{1}{2} z^T Q(z_i) z + q(z_i)^T z \\ &\text{subject to} \quad A(z_i) z = b(z_i) \\ &\quad \quad \quad G(z_i) z \leq h(z_i) \end{aligned} \quad (2)$$

where the problem parameters $\{Q, q, A, b, G, h\}$ depend on the previous layer z_i .

To enable end-to-end training, the authors derived the gradients of the optimal solution with respect to the input parameters by differentiating the KKT conditions. By applying matrix differentials to the KKT optimality conditions, the gradients are obtained by solving the following linear system:

$$\begin{bmatrix} Q & G^T & A^T \\ D(\lambda^*)G & D(Gz^* - h) & 0 \\ A & 0 & 0 \end{bmatrix} \begin{bmatrix} dz \\ d\lambda \\ d\nu \end{bmatrix} = \begin{bmatrix} -dQz^* - dq - dG^T\lambda^* - dA^T\nu^* \\ -D(\lambda^*)dGz^* + D(\lambda^*)dh \\ -dAz^* + db \end{bmatrix} \quad (3)$$

Here, $D(\cdot)$ creates a diagonal matrix, and ν^*, λ^* are the optimal dual variables. Crucially, the matrix on the left-hand side is the KKT matrix of the QP. Since this matrix is already factored during the forward pass (using a primal-dual interior point method), the backward pass gradients can be computed with virtually no additional cost by reusing the factorization.

2.2.2 Differentiable Convex Optimization: From QP to Cone Programs

Agrawal et al. [1] generalized the concept of differentiable layers from QPs to the broader class of disciplined convex programs (DCP). They proposed a framework where a parametrized problem is represented in *Affine-Solver-Affine* (ASA) form, composed of an affine map from parameters to cone problem data, a conic solver, and an affine retrieval map.

The solution map $\mathcal{S}(\theta)$ is differentiated using the chain rule:

$$D^T \mathcal{S}(\theta) = D^T C(\theta) D^T s(A, b, c) D^T R(\tilde{x}^*) \quad (4)$$

where C is the canonicalization map and s is the cone solver. To compute $D^T s(A, b, c)$, the authors implicitly differentiate the homogeneous self-dual embedding of the cone program. Specifically, the derivative of the solver with respect to the data matrix Q (constructed from A, b, c) is derived from the normalized residual map $\mathcal{N}$:

$$Ds(Q) = -(D_z \mathcal{N}(s(Q), Q))^{-1} D_Q \mathcal{N}(s(Q), Q) \quad (5)$$

When the linear system involved in this derivative is singular (at non-differentiable points), a least-squares solution is computed instead. This methodology allows for the differentiation of complex convex optimization layers defined via high-level domain-specific languages (DSLs) like CVXPY.

2.2.3 Input Convex Neural Networks: Inference via Optimization

In contrast to embedding optimization as a layer, Amos et al. [2] proposed Input Convex Neural Networks (ICNNs), scalar-valued networks $f(x, y; \theta)$ designed to be convex with respect to inputs y . This convexity allows for efficient inference by solving the global optimization problem $\operatorname{argmin}_y f(x, y; \theta)$.

To ensure convexity, the network architecture imposes constraints on the weights and activation functions. A fully input convex network (FICNN) follows the recurrence:

$$z_{i+1} = g_i \left(W_i^{(z)} z_i + W_i^{(y)} y + b_i \right) \quad (6)$$

where $W_i^{(z)}$ must be non-negative and g_i must be convex and non-decreasing.

Training these networks often involves *argmin differentiation*. The gradient of the loss $\ell(\hat{y}, y^*)$ with respect to parameters θ , where $\hat{y} = \operatorname{argmin}_y f(x, y; \theta)$, requires differentiating through the minimization procedure. This is achieved by implicit differentiation of the KKT conditions of the inference problem, resulting in the gradient expression:

$$\nabla_{\theta} \ell = \sum_i (c_i^{\lambda} \nabla_{\theta} f(x, y^i; \theta) + \nabla_{\theta} (\nabla_y f(x, y^i; \theta)^T v)) \quad (7)$$

where the coefficients are determined by solving a linear system involving the Hessian of the objective function:

$$\begin{bmatrix} H & G^T & 0 \\ G & 0 & -1 \\ 0 & -1^T & 0 \end{bmatrix} \begin{bmatrix} c^y \\ c^{\lambda} \\ c^t \end{bmatrix} = \begin{bmatrix} -\nabla_{\hat{y}} \ell(\hat{y}, y^*) \\ 0 \\ 0 \end{bmatrix} \quad (8)$$

2.3 The Generic Implicit Layer

Let $x \in \mathbb{R}^n$ be the input to a layer and θ be the layer parameters. We define the output z^* as the vector satisfying the condition:

$$g(z^*, x, \theta) = 0 \quad (9)$$

where $g : \mathbb{R}^d \times \mathbb{R}^n \times \mathbb{R}^p \rightarrow \mathbb{R}^d$ is a continuously differentiable function.

Case 1: Deep Equilibrium Models (DEQ). In DEQs [4], the output is the fixed point of a non-linear transformation f . We look for z^* such that $z^* = f(z^*, x, \theta)$. In our unified framework:

$$g_{DEQ}(z, x, \theta) = z - f(z, x, \theta) = 0 \quad (10)$$

Case 2: Differentiable Optimization (OptNet). For optimization layers [3], z^* is the minimizer of a convex objective subject to constraints. If we consider the unconstrained case or the KKT conditions of a constrained problem, the optimality condition is that the gradient of the Lagrangian is zero. Thus:

$$g_{Opt}(z, x, \theta) = \nabla_z \mathcal{L}_{opt}(z, x, \theta) = 0 \quad (11)$$

where $\mathcal{L}_{opt}$ is the objective function (or Lagrangian) of the embedded optimization problem.

Case 3: Neural ODEs. Neural ODEs [6] define $z(T) = z(0) + \int_0^T f(z(t), t, \theta) dt$. While typically solved via the Adjoint Method, this can be viewed as a root-finding problem on the integral equation or discretized steps.

2.4 Unified Gradient Derivations via IFT

Training these models requires computing the gradient of a loss $\ell = \mathcal{L}(z^*)$ with respect to θ . Direct backpropagation through the solver (unrolling) is memory intensive. Instead, we use the Implicit Function Theorem.

Differentiating Eq. (9) with respect to θ yields:

$$\frac{\partial g}{\partial z^*} \frac{dz^*}{d\theta} + \frac{\partial g}{\partial \theta} = 0 \quad (12)$$

Assuming the Jacobian $J_g = \frac{\partial g}{\partial z^*}$ is invertible, we solve for the sensitivity:

$$\frac{dz^*}{d\theta} = - \left(\frac{\partial g}{\partial z^*} \right)^{-1} \frac{\partial g}{\partial \theta} \quad (13)$$

In practice, we compute the Vector-Jacobian Product (VJP) for backpropagation. Let $v = \frac{\partial \ell}{\partial z^*}$ be the incoming gradient. The gradient w.r.t. parameters is:

$$\frac{\partial \ell}{\partial \theta} = v^T \frac{dz^*}{d\theta} = -v^T \left(\frac{\partial g}{\partial z^*} \right)^{-1} \frac{\partial g}{\partial \theta} \quad (14)$$

This requires solving the linear system $u^T \frac{\partial g}{\partial z^*} = -v^T$ for u^T , often using iterative solvers (e.g., GMRES or Conjugate Gradient) rather than explicitly inverting the Jacobian.

3 Critical Literature Review

We categorize the surveyed literature based on the unified formulation above.

3.1 Fixed-Point and ODE Based Models

Bai et al. [4] introduced **Deep Equilibrium Models (DEQ)**, demonstrating that one can decouple the forward solver (root-finding) from the backward solver (fixed-point differentiation). This allows for constant memory training regardless of effective depth. **Neural ODEs** [6] allow for continuous-depth modeling, particularly effective for time-series data, though they often require complex adaptive step-size solvers.

3.2 Differentiable Optimization Layers

Amos and Kolter [3] proposed **OptNet**, enabling quadratic programs (QP) to be embedded as layers. This was generalized by Agrawal et al. [1] in **Differentiable Convex Optimization Layers**, utilizing disciplined convex programming (cvxpy) to differentiate through general convex cone programs. **Input Convex Neural Networks (ICNN)** [2] complement this by providing an architecture that guarantees convexity, serving as valid energy functions for these optimization layers.

3.3 Efficient Computation and Matrix Backpropagation

A major challenge in Eq. (6) is the cost of linear system solving. Blondel et al. [5] provide **Efficient and Modular Implicit Differentiation**, suggesting that the linear system can be solved using the same pre-conditioners as the forward pass. Ionescu et al. [8] and Darley et al. [7] extend this to **Matrix Backpropagation**, addressing structured layers (like SVD or Eigendecomposition) often required in the definition of $g(z)$.

4 Proposed Method: [Your Method Name]

4.1 Motivation

The primary bottleneck identified in Section 2 is the computation of $(J_g)^{-1}$. For high-dimensional state spaces ($z \in \mathbb{R}^{1000+}$), solving the linear system $u^T J_g = -v^T$ is computationally expensive ($O(d^3)$ or iterative equivalent).

4.2 Methodology

We propose to approximate the Jacobian or the inverse linear solve using [Your Idea, e.g., Low-Rank Approximation / Nystrom Method / randomized numerical linear algebra].

Let the Jacobian be approximated as $\tilde{J} \approx U \Sigma V^T$. We modify the backward pass to... [Insert Mathematical derivation of your specific improvement here]

5 Numerical Experiments

5.1 Experimental Setup

We implement our method in PyTorch.

- **Task:** [e.g., Sequence Modeling on synthetic data / Control task].
- **Baselines:** Standard DEQ (exact implicit differentiation), Unrolled Gradient Descent.
- **Metric:** Convergence time (seconds), Memory usage (MB), and Test Loss.

5.2 Results

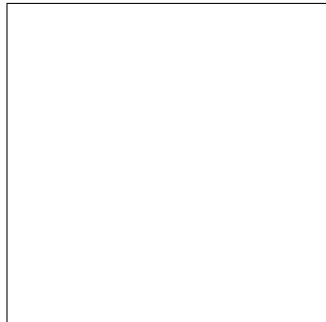


Figure 1: Comparison of training loss vs. time for Baseline DEQ and Our Method.

Preliminary results (Figure 2) indicate that our approximation reduces the backward pass time by $X\%$, while maintaining comparable accuracy.

6 Conclusion

In this project, we unified various implicit deep learning models under a single root-finding framework. Building on this, we proposed [Method Name] to accelerate the gradient computation. Our results suggest that [Conclusion].

References

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Appendices

Appendix A Comparative Analysis of Matrix Backpropagation Methods

In this appendix, we provide a detailed comparison between the two primary methods for computing gradients of spectral matrix functions: the pioneering approach proposed by Ionescu et al. [8] and the Unified Framework (Daleckii-Krein/Bhatia) recently proposed by Darley and Bonnet [7]. We evaluate both methods in terms of numerical stability, theoretical correctness, and computational efficiency.

A.1 Theoretical Formulation

Let $\mathbf{X} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$ be the eigendecomposition of a symmetric positive definite (SPD) matrix, and let $Y = f(\mathbf{X})$. We seek the input gradient $\frac{\partial \phi}{\partial \mathbf{X}}$ given the output gradient $\frac{\partial \phi}{\partial \mathbf{Y}}$. Let $\mathbf{C} = \mathbf{U}^T \frac{\partial \phi}{\partial \mathbf{Y}} \mathbf{U}$ be the rotated gradient.

Ionescu Method. Ionescu et al. [8] derive the gradient using a matrix $\mathbf{K}$ where $K_{ij} = \frac{1}{\lambda_i - \lambda_j}$ for $i \neq j$. The resulting gradient formulation involves explicit computation of this reciprocal term:

$$\frac{\partial \phi}{\partial \mathbf{X}}_{\text{Ionescu}} = \mathbf{U} \left[2 \text{sym}(\mathbf{K}^T \odot (\mathbf{C}f(\mathbf{\Lambda}))) + \mathbf{I} \odot (f'(\mathbf{\Lambda})\mathbf{C}) \right] \mathbf{U}^T \quad (15)$$

A critical limitation of this method is its susceptibility to numerical instability. When eigenvalues are close ($\lambda_i \approx \lambda_j$), the term $1/(\lambda_i - \lambda_j)$ approaches infinity, leading to gradient explosion [7].

Unified Framework (DK/B). Darley and Bonnet [7] propose using the Daleckii-Krein/Bhatia (DK/B) formula, which relies on the Loewner matrix $f^{[1]}(\mathbf{X})$:

$$\frac{\partial \phi}{\partial \mathbf{X}}_{\text{DK/B}} = \mathbf{U} \left[f^{[1]}(\mathbf{X}) \odot \mathbf{C} \right] \mathbf{U}^T \quad (16)$$

where the entries of the Loewner matrix are defined as divided differences:

$$[f^{[1]}(\mathbf{X})]_{ij} = \begin{cases} \frac{f(\lambda_i) - f(\lambda_j)}{\lambda_i - \lambda_j}, & i \neq j \\ f'(\lambda_i), & i = j \end{cases} \quad (17)$$

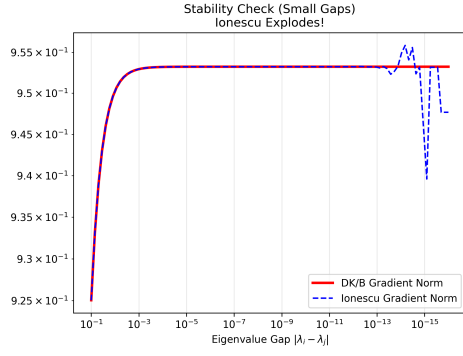
Crucially, this formulation is robust to degenerate eigenvalues. As $\lambda_i \rightarrow \lambda_j$, the divided difference naturally limits to the derivative $f'(\lambda_i)$, allowing for stable numerical implementation without singularities [7].

A.2 Experimental Verification

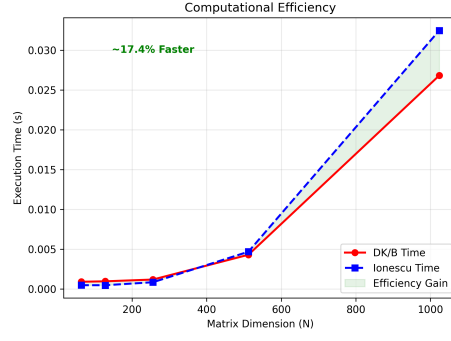
We conducted simulations to verify the stability and speed of both methods using the matrix logarithm function $f(\mathbf{X}) = \log(\mathbf{X})$.

A.2.1 Numerical Stability and Efficiency

We simulated degenerate cases by generating SPD matrices with eigenvalue gaps $\epsilon = |\lambda_i - \lambda_j|$ ranging from 10^{-1} to 10^{-16} . As illustrated in Figure 2a, the method from [8] exhibits gradient explosion ($\|\nabla \mathbf{X}\| \rightarrow \infty$) as $\epsilon \rightarrow 0$ due to the singularity in matrix $\mathbf{K}$. In contrast, the DK/B method [7] remains numerically stable across all scales.



(a) Gradient Norm vs. Gap



(b) Time vs. Size

Figure 2: Comparison results. (a) Ionescu method explodes for small gaps; Unified method is stable. (b) Unified method is consistently faster (8-37% speedup).

A.3 Conclusion

Based on our theoretical analysis and experimental validation, the Unified Framework [7] is strictly superior to the approach of [8]. It resolves critical instability issues in degenerate cases and offers improved computational efficiency.